Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/06/2020
PI Name:	Rice, Kelly Ph.D.		
Project Title:	Probing the Effect of Simulated Microgravity on the Pathogenic Potential of Cariogenic Streptococcus mutans		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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Zip Code:	32611-0700	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	07/25/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	August 2020: No key personnel changes to report.		
COI Name (Institution):	Shields, Robert Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC18K1463		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Streptococcus mutans is a member of the human oral microbiota and a major etiological agent of dental caries (cavities) due to its ability to form biofilm, adapt to oral cavity stresses, and rapidly ferment carbohydrates. These characteristics promote its predomination of dental plaque biofilm and combined with repeated cycles of fermentative acid production, leads to tooth demineralization and cavity formation. Using High Aspect Rotating Vessels (HARVs) in the low-shear modelled microgravity (LSMMG) model, we previously demonstrated that S. mutans displays several physiological changes in response to simulated microgravity, including altered cell aggregation, enhanced sensitivity to oxidative stress, and global alterations in gene expression. Furthermore, simulated microgravity exposure has been shown by others to cause increased mandibular and alveolar bone loss and decreased saliva flow, factors that could predispose astronauts to caries. Therefore, the goal of this application is to characterize the physiological and genetic response of S. mutans to simulated microgravity, focusing specifically on biofilm development, stress resistance, and identifying key gene(s) and/or the core transcriptomic response that favors S. mutans survival during microgravity growth. This will be accomplished by three objectives: (1) S. mutans will be cultured in biofilm-promoting media under simulated microgravity conditions using the LSMMG and random-positioning machine (RPM) ground-based models of growth. Changes in cell aggregation, stress resistance, and gene expression profiling (via RNAseq) will be assessed with respect to normal gravity control cultures, and these results will be compared using bioinformatics and statistical techniques. This analysis will allow us to make more refined predictions as to which physiological and gene expression changes are likely to occur in S. mutans during spaceflight, and how this may impact its pathogenic potential; (2) Changes in S. mutans biofilm development in response to simulated microgravity will be studied using 2 ground-based models (LSMMG, RPM). S. mutans will be assessed in these models for biofilm viability and ultrastructure, exopolysaccharide production, and for its ability to compete with non-cariogenic oral streptococci; (3) A S. mutans Tn-seq library will be screened for mutants that lose competitive fitness during LSMMG and RPM-based simulated microgravity growth. This will be accomplished by using high-throughput DNA sequencing of pooled batches of Tn mutants pre- and post-HARV growth under both simulated microgravity and normal gravity conditions. This research is directed at understanding the physiological and genetic response of S. mutans to microgravity, especially with respect to multicellular behavior (biofilm formation), stress resistance, and the genes that favor S. mutans survival during microgravity growth. Therefore, this application directly addresses guiding questions MB-1 (What underlying genetic, molecular and biochemical processes are influenced by the space environment?), MB-3 (How does spaceflight influence microbial biofilm formation and microbial community development?), and MB-4 (What are the mechanisms that effect changes such as the altered virulence or altered drug resistance observed in some organisms during spaceflight?) of the 2016-2025 Space Biology Science Plan, each of which aligns with Priority P1 (multigenerational studies of microbial populations) of the National Research Council's (NRC) 2011 Decadal Survey Report.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Collectively, the research proposed in this application is directed at understanding the physiological and genetic response of S. mutans to simulated microgravity, especially with respect to multicellular behavior (biofilm formation), stress resistance, and the underlying genetics that promote S. mutans fitness and growth. It is anticipated that the collective data generated from these three objectives will allow us to make better predictions as to which physiological and gene expression changes are likely to occur in S. mutans during spaceflight, and how this may impact its disease-causing potential and effect on astronaut health. The data generated in objectives 1-3 of this application will also contribute to our understanding of the molecular mechanisms by which S. mutans and other oral bacteria contribute to dental caries (cavities), since the low-shear force experienced during HARV simulated microgravity growth is very similar to that of saliva flow in the oral cavity.
Task Progress:	Overall Task Progress: In this second funding period, 2 x 5-day field trips to the Microgravity Simulation Support Facility (MSSF) at Kennedy Space Center (KSC) were completed, during which time data and samples were collected for S. mutans Random Positioning Machine (RPM) experiments, as well as further optimization of our biofilm RPM model. We have also completed Rotary Cell Culture System (RCCS) HARV experiments for simulated microgravity and normal gravity controls. The bulk of our research progress has been focused on Objective 1 during this funding period. The graduate student assigned to this project has presented her research at two conferences: 1. Poster presentation: “Effects of simulated microgravity on Streptococcus mutans physiology”, 35th Annual Meeting of the ASGSR, Denver, CO, Nov. 20-23, 2019. 2. Oral presentation: "Effects of simulated microgravity growth on Streptococcus mutans physiology", McKnight Doctoral Fellowship Annual Meeting, Tampa, FL, February 21-22, 2020. Our research productivity in year 2 of this award was delayed due to COVID shutdown and restart of our research lab between early March and late June, 2020. We have also added a part-time research technician to this project to assist in completing the remainder of experiments in Year 3 of this project. Research progress has also been made as follows: • Data and samples were collected from S. mutans Random Positioning Machine (RPM) mid-exponential phase and early stationary phase cultures for growth measurements, cell pellets for RNASeq, and stress assays. Growth data, stress assays, and cell pellets for RNAseq were also collected in our lab at University of Florida (UF) using RCCS HARV mid-exponential and early stationary phase S. mutans cultures for simulated microgravity and normal gravity controls. Cell pellets for RNASeq, growth data, and acid tolerance stress assays have also been collected for corresponding S. mutans control tube cultures. • Data analysis for growth data in all 4 conditions, hydrogen peroxide stress assays, and acid tolerance assay data have been completed. There appears to be a growth-phase dependent effect on resistance to hydrogen peroxide (RCCS, RPM) and acid stress (RPM, control tubes), but no statistically significant differences were observed between simulated microgravity and normal gravity/control cultures. • Bioinformatics analysis of our previously-published S. mutans RNAseq dataset (RCCS microgravity vs. normal gravity comparison at stationary phase growth) has identified potential oxidative stress genes that were differentially regulated between these two growth conditions. We have created a S. mutans genetic mutant in one of these genes (dpr), and its sensitivity to oxidative stress has been confirmed. We will test this mutant for its growth properties and stress resistance phenotypes using the RCCS model in Year 3 of this project.

	• Further work with the RPM growing S. mutans biofilm cultures revealed that our current fluorescent reporter strains are not consistently bright enough for use in imaging RPM biofilms. Therefore, we are engineering our S. mutans and S. gordonii strains with improved fluorescent reporter plasmids. • Our Co-Investigator has S. mutans TnSeq libraries we can use for TnSeq RCCS and RPM experiments. COVID delayed planned productivity on this experiment, thus the bulk of this work will be performed in year 3 of this project. The research technician newly added to this project will primarily assist with these studies.
Bibliography Type:	Description: (Last Updated: 02/19/2026)
Abstracts for Journals and Proceedings	Davis K, Winkel C, Rice KC. "Effects of simulated microgravity growth on Streptococcus mutans physiology (SP47)." Undergraduate/Graduate Poster Session, 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019